

Resolving diachronic linguistic changes in Biblical Hebrew: calibrated ranges for the Pentateuchal sources, with cross-linguistic replication in Ancient Greek

Aaron Adair^{1*}

¹ Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, USA

* Corresponding author: adairaar@mit.edu

Abstract

The linguistic dating of ancient texts from internal morphosyntactic evidence is central to biblical studies and classical philology, but how finely such a text can actually be placed in time has not been measured directly. Reported accuracies vary by two orders of magnitude, in part because validation designs in this area frequently supply a text's expected date to the model that is then asked to recover it. The present paper measures it. Every out-of-sample result uses an agnostic prior; feature screening, standardisation and model fitting are performed inside each cross-validation fold; and significance is established by permutation of the date labels through the entire pipeline rather than by parametric assumption. Two model families are fitted and both are reported throughout. Applied to 25 units of the BHSA database spanning 760–167 BCE, the analysis yields a sharply bounded positive result. Hebrew morphosyntax carries a genuine diachronic signal: the models order text pairs chronologically with 69.8% and 67.8% accuracy across 295 pairs, against a permutation null of 47.6% and 43.6% ($p = 0.022, 0.029$), and leave-one-out Spearman correlations between true and predicted dates are +0.58 and +0.49 ($p = 0.016, 0.024$). That signal does not, however, support absolute dating: leave-one-out mean absolute error is 121–156 yr against 137 yr for the trivial predictor that assigns every text the corpus mean date, and four-way period assignment reaches only 40–48%. Holdout accuracies of order 10–30 yr, of the kind reported for models of this class, arise when a held-out text is dated under a prior centred on its own scholarly date: across the units here the linguistic evidence supplies a median 12% of posterior precision, and under 6% for the most tightly anchored texts. Applied to the Pentateuchal sources with distribution-free conformal intervals, the framework supports one substantive conclusion — the Priestly, Deuteronomic and JE composites are post-exilic with probability ≥ 0.92 under both families — while declining to place them within a particular post-exilic period. That probability is itself validated: on the dated corpus, confident post-exilic calls are correct on 17 of 18 instances across the two families, with positive Brier skill against the base rate ($p = 0.005, 0.014$). The Song of the Sea and Song of Deborah emerge as the two earliest of nineteen undated units in both families without having been trained on. Code and corpora are publicly available (see Data Availability).

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH; also called Standard Biblical Hebrew, SBH) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. Despite this rich empirical tradition, linguistic dating almost never produces calibrated probability estimates; scholars identify features as “archaic” or “late” and assign texts to qualitative categories, but the resulting claims carry no quantified uncertainty.

A recent contribution by Faigenbaum-Golovin et al. [5] demonstrated that statistical analysis of lemma n-gram frequencies can reliably distinguish among the three major scribal corpora of the Hebrew Bible — the old layers of Deuteronomy (D), the Deuteronomistic History (DtrH), and the Priestly source (P) — achieving approximately 84% chapter-level attribution accuracy on a 50-chapter benchmark, with interpretable discriminating-lemma lists linking computational results to traditional philological categories. That study establishes the viability of quantitative computational methods for biblical Hebrew scholarship and demonstrates that linguistically distinct scribal schools leave detectable statistical fingerprints even in relatively short texts. Its focus, however, is *authorship attribution*: given a passage of disputed origin, which known scribal milieu produced it? The present paper addresses the complementary but methodologically distinct question of *diachronic dating*: not “which school wrote it?” but “when was it written?” These are related but separable questions that demand different statistical frameworks, different feature sets, and different validation strategies, as detailed below.

This paper addresses two methodological gaps that authorship-attribution approaches leave open. First, no existing probabilistic framework provides date posteriors with credible intervals for Biblical Hebrew texts. Qualitative assessments of individual features are genuinely informative, but they cannot be combined into a single coherent posterior without a model that propagates feature-level evidence into a joint probability distribution over dates. Second, no principled framework separates the temporal signal from two well-known confounds: *archaism* (a late composer deliberately employing archaic vocabulary and morphology) and *genre* (legal or narrative register producing feature distributions that mimic either earlier or later stages of the language independently of composition date).

The present framework contributes the following:

1. A morphosyntactic feature extraction pipeline operating on the BHSA/ETCBC database [6, 7], producing 35 features across three tiers (lexical, morphological, and clause-level syntactic), selected by Spearman correlation ($p < 0.30$) with 100% LOO sign-consistency.
2. The MLE-MVN Bayesian dating model: per-feature OLS regression, Tikhonov-regularized residual covariance, and posterior over a 500-point date grid with explicit uncertainty quantification.
3. Prior sensitivity analysis distinguishing data-driven from prior-dominated date estimates, and a decomposition of posterior precision that quantifies how much of a reported date is contributed by the linguistic evidence rather than by the prior.
4. A leave-one-out protocol in which feature screening, standardisation and model fitting are performed inside each fold, and every out-of-sample unit is dated under an agnostic prior.
5. Permutation inference: the date labels are shuffled and the entire pipeline re-run, so the null distribution absorbs selection effects.
6. Distribution-free conformal prediction intervals calibrated on leave-one-out residuals, replacing parametric intervals whose empirical coverage is roughly three-quarters of nominal.

The results are bounded in both directions. Hebrew morphosyntax carries a genuine diachronic signal: under leakage-free cross-validation the models order text pairs chronologically with 69.8% and 67.8% accuracy across 295 pairs against a permutation null near 45% ($p = 0.022, 0.029$), and rank correlations between true and predicted date reach +0.58 and +0.49 in two independently specified model families. The same analysis shows that this signal does not support absolute dating: leave-one-out error of 121–156 yr

must be read against 137 yr for the trivial predictor assigning every text the corpus mean date, and four-way period assignment reaches 40–48%. We further show that holdout accuracies of order 10–30 yr, of the kind reported for models of this class, arise when a held-out text is dated under a prior centred on its own scholarly date; for the most securely anchored texts the linguistic evidence contributes under 6% of posterior precision. Applied to the Pentateuchal sources under corrected conditions, the framework establishes that the Priestly, Deuteronomic and JE composites are post-exilic with probability ≥ 0.92 under both model families, while declining to locate them within a particular post-exilic period.

The paper is organized as follows. The Background section reviews the CBH/LBH distinction and the methodological debates surrounding linguistic dating. The Data section describes the training corpus and feature extraction. The MLE-MVN section presents the generative dating model and a ridge regression fitted alongside it. The Validation section documents the prior-leakage failure mode, sets out the leave-one-out protocol, and reports what the signal does and does not support. The Results section applies the corrected framework to the undated units. The Discussion section addresses limitations, extensibility, and the relationship to authorship-attribution approaches.

Background: diachronic Biblical Hebrew

The CBH/LBH distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel-Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2, 3, 8–10]. At the lexical level, the first-person singular pronoun shifts from *ankī* (CBH) to *anī* (LBH); the relative pronoun *ašer* gives way to the enclitic *šē-*; the negator *lō'* is supplemented by and eventually partly replaced by *'ēn*. At the morphological level, the wayyiqtol (narrative imperfect) declines in frequency while the *hāyāh* + participle periphrastic progressive increases. At the syntactic level, infinitive-absolute constructions rise, and the overall sentence structure shifts toward the analytic patterns associated with Aramaic influence [11, 12].

The consensus view [1, 13–15] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption while formally modeling and partially testing the alternatives.

The diachronic versus dialectal debate

Young, Rezetko, and Ehrensverd [11] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (Dead Sea Scrolls, ~150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern interpreted as literary register rather than temporal reality.

Hurvitz and the diachronic mainstream [8] counter that the dialectal alternatives require additional auxiliary hypotheses; and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

For the present paper, the question is not which side of this debate is correct but whether the empirical evidence can distinguish between them. The framework operationalizes the distinction rather than adjudicating it on a priori grounds. One natural operationalization is to ask whether clause-level syntactic patterns, which fall below the level of deliberate stylistic choice, agree with a date derived from lexical and morphological features that a competent imitator could manipulate; systematic disagreement would be a falsifiable indicator of archaism. We implemented this test and report under Results: dating the undated units that it did not function on the present corpus, flagging no features on any unit. The archaism question therefore remains open here, and the distinction between an early text and a skilled archaizing one is one this analysis cannot draw.

The circularity problem

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. Three safeguards are adopted.

First, the training corpus is restricted to prophetic books with *independent* historical anchoring via synchronisms with datable rulers in Assyrian, Babylonian, and Persian king-lists. Torah texts are excluded from training entirely, so all Torah date estimates are genuine out-of-sample predictions.

Second, a held-out validation set (oracle Jeremiah chapters) was withheld from all feature selection and training. Its correct recovery by the model (see the Validation section) provides empirical evidence against the circularity objection.

Third, the precision decomposition reported under Validation explicitly quantifies how much of a reported date is determined by the scholar-specified prior versus the linguistic data.

Beyond these three safeguards, the choice of the prophetic corpus as the training base deserves explicit justification, and the *direction* of any residual misdating error is methodologically important.

Why the prophetic corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — “the fifth year of King Jehoiachin’s captivity” (Ezek 1:2), “the sixth year” (8:1), and so forth — which can be correlated with Babylonian administrative texts that independently date Jehoiachin’s deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are independently documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates are derived entirely from literary-critical arguments, which is precisely the kind of circularity the present study seeks to avoid. The prophetic corpus therefore represents the most defensible choice for a training set whose dates are genuinely independent of the linguistic features under analysis.

Asymmetric error and conservative upper limits. If the accepted dates for the prophetic training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference (the superscriptions and Ezekielian dates provide firm termini post quos), but they could in principle be *younger* — composed or reaching final form some time after the events they describe. This asymmetry has a direct consequence for the date estimates produced by the model. A training corpus that is misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis; unknown texts dated by reference to that calibration would also be estimated as too early, meaning their true dates would be *later* than the model reports. In other words, if the prophetic training dates carry any systematic error, that error propagates in a single direction: all model-derived dates shift toward greater recency, not greater antiquity. The model’s output for a text such as the Song of the Sea — whose predictive mass under both model families is the earliest of any undated unit — would be pushed later still if the training corpus were found to be younger than traditionally assumed. The dates reported in this study should therefore be understood as *conservative upper bounds on antiquity*: a text dated here to 400 BCE may be more recent, but the linguistic evidence as calibrated against the prophetic corpus provides no support for a date earlier than that estimate.

Prior computational work: authorship attribution versus diachronic dating

Computational approaches to textual attribution and stylometric dating have a substantial prior history, from Mosteller and Wallace’s Bayesian analysis of the Federalist Papers [16] through the stylometric tradition surveyed by Holmes [17] and the function-word methods of Burrows [18]. Among work applied specifically to the Hebrew Bible, Faigenbaum-Golovin et al. [5] provide the most methodologically rigorous prior computational treatment. Their Higher Criticism (HC) discrepancy method [5] tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null hypothesis that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, DtrH, and P, the method achieves 84% attribution accuracy in leave-one-out validation and yields interpretable

discriminating-lemma lists that directly engage traditional philological categories. The statistical optimality properties of the HC statistic (sensitivity to rare but consistent frequency deviations across a large vocabulary) make it well-suited to its task.

The present framework is designed to answer a different question. Five methodological differences follow from that distinction.

Continuous date posteriors versus categorical corpus assignment. The HC method assigns each text to the most likely of three corpora. Because D and DtrH are both dated to the late monarchic or exilic period, two texts composed at 650 and 590 BCE receive identical labels if both score closest to DtrH; no intra-school temporal resolution is possible. The models used here instead compute a posterior probability distribution over a 500-point continuous date grid, recovering whatever temporal resolution the data support and quantifying uncertainty explicitly via credible intervals.

Morphosyntactic features versus raw lemma frequencies. Faigenbaum-Golovin et al. operate on raw lemma and lemma n-gram frequencies — a bag-of-words representation in which content words (e.g., *gold* in Tabernacle chapters, *king* in regnal narratives) dominate the signal. Content-word frequencies are sensitive to *topic* and *genre* as much as to *authorship*, and they carry little temporal signal for cross-genre comparisons. The 35 features used in this study are drawn from a century of diachronic Hebrew linguistics scholarship [1–3, 11] precisely because they are sensitive to *period* rather than *topic*: the *ank̄*→*an̄* pronoun shift, the negator transition, the wayyiqtol decline, and the infinitive-absolute rate change systematically across time and resist topical or generic influence. This distinction is especially important for cross-genre comparisons: bag-of-words features cannot compare poetic or legal texts against narrative texts without large genre confounds.

Archaism diagnostic. The HC method provides no mechanism for distinguishing a genuinely ancient text from one composed in a deliberately archaic register. A Persian-period author who archaizes vocabulary to D/DtrH norms will be assigned to DtrH regardless of actual composition date. The resistant/full-model divergence introduced in Section 7 detects this pattern directly: surface features under conscious compositional control (lexicon, morphology) and deep clause-level syntactic patterns that operate below deliberate stylistic choice are evaluated independently, and systematic divergence between their date estimates constitutes a quantified, falsifiable indicator of archaism. This diagnostic has no counterpart in the HC framework.

External holdout validation. Faigenbaum-Golovin et al. validate via leave-one-out and *k*-fold cross-validation within the three training corpora. This confirms internal consistency but cannot test whether the method recovers *correct* calendar dates, because the ground-truth labels are scribal-school assignments rather than independently anchored absolute dates. The present framework is validated against texts anchored to external historical events — oracle Jeremiah chapters to the Neo-Babylonian period [19], Haggai to 520 BCE (Persian regnal date), Daniel Hebrew chapters to 167 BCE (Maccabean crisis) — providing a genuinely independent test that the temporal signal is real rather than self-referential.

Cross-linguistic generalization. A 63-text Ancient Greek corpus spanning 440 BCE–360 CE was re-analysed under the identical protocol (Fig 1B,C). Holdout recoveries “within ~30 yr” of the kind reported for Greek corpora of this type are produced by the same prior-leakage design documented under Validation. What the corrected analysis supports is weaker but genuinely cross-linguistic: pairwise ordering accuracy of 60.3% and 58.9% against a permutation null near 40% ($p = 0.037, 0.050$), with leave-one-out rank correlations of +0.36 and +0.27. The ordinal signal replicates in an unrelated language at comparable significance, and that is the generalization claim the data support.

The negative result replicates more sharply still. Greek leave-one-out MAE is 294.5 yr (generative) and 205.4 yr (ridge) against 210.9 yr for the constant predictor: one family is substantially worse than a constant and the other improves on it by five years ($p = 0.24$). Holdout errors under the corrected design are 339.0 and 171.2 yr, against 8.7 yr under the leaky design. That this occurs at $n = 63$ over an 800-year span — where a small corpus cannot be blamed — indicates the limitation is intrinsic to the approach rather than a consequence of sample size.

These five differences are differences of purpose rather than deficiencies in either approach. Authorship attribution and diachronic dating ask related but separable questions; the ideal analysis of any disputed text would apply both. The HC output (scribal-school membership) can in principle be used to set register priors, and conversely the date estimates reported here constrain the plausible compositional settings for HC

attribution decisions. This complementarity is taken up in the Discussion.

Data: corpus and feature extraction

The BHS database

All Hebrew text analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHS) database, version 2021, produced by the Eep Talstra Centre for Bible and Computer (ETCBC) at Vrije Universiteit Amsterdam [6, 7, 20]. BHS provides morphological tagging at the word level, phrase-level function annotation, clause boundary segmentation, and lexeme coding for the entire Hebrew Bible. Data are accessed programmatically via the Text-Fabric Python library, enabling reproducible extraction without manual annotation. No Aramaic chapters are included in training or testing.

Training corpus

The dated corpus comprises 25 units spanning 760–167 BCE and 146,059 tokens: 23 training units and two units held out of both models (Table 1). Date anchoring uses explicit synchronisms with datable ancient Near Eastern events wherever these are available. Amos 1:1 places that book in the reigns of Uzziah and Jeroboam II (~760 BCE); Zechariah 1–8 and Haggai carry explicit Darius regnal dates (518 and 520 BCE); Ezekiel supplies thirteen regnal dates checkable against Babylonian records; Daniel’s Hebrew chapters (1, 8–12) are anchored to 167 BCE by the Maccabean crisis. Daniel is restricted to those Hebrew chapters; the Aramaic chapters (2–7) are a different language and are excluded.

Not every unit is anchored this securely, and the corpus does not pretend otherwise. Where a date rests on literary-critical rather than synchronistic grounds, σ is widened accordingly: Joel and Zechariah 9–14 carry $\sigma = 150$ yr, Ecclesiastes 120 yr, Isaiah 56–66 100 yr, and Obadiah 80 yr, against 10–30 yr for the synchronistically anchored texts. The widening is deliberate: a narrow prior on a literary-critical date imports the very assumption the analysis is meant to test, and Validation quantifies what such an assumption is worth in this setting.

All analyses are leave-one-out over the full set of 25 dated units, so every unit is held out in turn and no unit contributes to its own estimate. The corpus therefore has no separate holdout stratum. Whole-book Jeremiah is excluded entirely and replaced by its oracular stratum (chs. 1–6, 8–10, 12–16, 19–20, 22–23, 30–31, 46–51), since the whole book contains that stratum; the Deuteronomistic prose stratum (chs. 7, 11, 17–18, 21, 24–29, 32–45, 52) is treated as an undated target and never trained on.

A note on what “held out” must mean. In the analyses reported below, a held-out unit contributes nothing to feature screening, to standardisation, to regression fitting, or to its own prior. The last of these is not a formality. A model that receives a text’s scholarly date as a prior mean will return approximately that date whatever the text contains, and will do so most emphatically for the texts whose dates are best established — that is, precisely the texts on which a validation exercise appears most convincing. Validation quantifies the effect on this corpus. All out-of-sample results in this paper use an agnostic prior.

S5 Table lists word counts for all units, including the Documentary Hypothesis source composites, which are excluded from both training and validation and are dated solely from the model.

Feature extraction

Features are extracted at three tiers that differ in their susceptibility to deliberate stylistic manipulation (Table 3).

Tier 1 (lexical, 19 features). Rates per 1,000 words of specific function words, particles, demonstratives, and prepositions. Examples: the *ašer/še* ratio; the *ankī/anī* ratio; the absolute rate of the particle *gam*; the rate of the accusative marker *et*. Tier 1 features most directly capture the CBH/LBH lexical axis but are also most susceptible to deliberate archaism, because a competent composer can deliberately choose archaic function words.

Tier 2 (morphological, 12 features). Verb stem fractions (qal, piel, hif, niphal as fractions of all verbs); verb form fractions (wayyiqtol, qatal, yiqtol, participle, imperative fractions); absolute noun ratio;

Table 1. Dated corpus: 23 training units and 2 held-out units. Date (BCE) is the scholarly consensus midpoint used as the training label; σ is the accompanying uncertainty, set wider on every text whose date rests on literary-critical rather than synchronistic grounds. Register: S = SBH (Standard Biblical Hebrew, equivalent to CBH), T = Transitional, L = LBH. Word counts are token counts from the BHS extraction used for all feature rates. ^bHeld out of both models, and excluded from all feature screening. Whole-book Jeremiah is excluded from training entirely, because it contains the Jeremiah oracle holdout.

Unit	Register	Date (BCE)	σ (yr)	Words
<i>Training</i>				
Amos	S	760	20	2,780
Hosea	S	745	25	3,146
Micah	S	725	25	1,895
Isaiah 1–39	S	720	30	13,504
Nahum	S	650	25	746
Zephaniah	S	630	20	1,037
Habakkuk	S	605	20	897
Obadiah	T	585	80	392
Ezekiel	T	580	20	26,182
Lamentations	T	570	50	1,945
Isaiah 40–55	T	545	20	5,738
Zechariah 1–8	T	518	10	2,484
Malachi	T	460	25	1,187
Joel	T	400	150	1,318
Jonah	T	400	60	985
Isaiah 56–66	T	400	100	3,689
Ezra	L	380	60	5,268
Nehemiah	L	380	60	7,842
Zechariah 9–14	L	350	150	1,987
Chronicles	L	330	40	35,330
Esther	L	300	60	4,621
Ecclesiastes	L	250	120	4,233
Daniel (Heb.)	L	167	10	3,437
<i>Held out of both models</i>				
Jeremiah oracle ^b	S	605	30	14,539
Haggai ^b	T	520	10	877
Total				146,059

pronominal suffix rate. These are harder to archaize wholesale but remain subject to word-level editorial manipulation.

Tier 3 (clause/phrase, 4 features). Four clause-level features, the least susceptible of the three tiers to deliberate manipulation: the infinitive-construct fraction (`frac_inf`), the fronted-clause fraction (`frac_fronted`), the null-subject fraction (`frac_null_subj`), and the waw-qatal fraction (`frac_wqt1_wayq`). These measure syntactic templating patterns that are acquired below the level of conscious stylistic choice and are therefore most resistant to deliberate manipulation.

Feature selection. Spearman rank correlation with date is computed for each candidate feature against the 22-text training set. Features are retained at a permissive significance threshold of $p < 0.30$: with only $N = 22$ training texts, a strict Benjamini-Hochberg FDR correction [21] at $\alpha = 0.05$ would retain only ~ 10 features, which is insufficient to capture the multi-dimensional CBH/LBH signal documented over a century of Hebrew linguistics scholarship. The $p < 0.30$ threshold is paired with a leave-one-out (LOO) robustness filter: a feature is retained only if its Spearman ρ preserves sign in $\geq 68\%$ of LOO resamplings (the 68% threshold corresponds to one standard deviation of a binomial sign-consistency test, providing a principled stopping criterion). In practice, all 35 features surviving the $p < 0.30$ screen achieve 100% LOO sign consistency (LOO fraction = 1.00), well above the 68% floor, confirming that the selected features carry a

robust diachronic signal. This yields a final set of **35 features**. Table 2 lists the ten features with the largest $|\rho|$; the complete set is provided in S1 Table. OLS slopes ($\hat{\beta}_j$) and their confidence intervals for all 35 features are shown in S6 Fig.

Table 2. Top-10 features by Spearman $|\rho|$ with date. All have LOO fraction = 1.00. Sign of ρ indicates direction: negative = feature declines with date (more CBH-like in older texts). Full 35-feature list in S1 Table.

Rank	Feature	Tier	Spearman ρ	Direction
1	<i>ankī/anī</i> pronoun ratio	1	−0.94	Declines
2	Wayyiqtol rate	2	−0.91	Declines
3	<i>ašer</i> relative rate	1	−0.89	Declines
4	<i>še-</i> relative fraction	1	+0.87	Rises
5	<i>lo'/'en</i> negator frac.	1	+0.85	Rises
6	Qatal rate	2	+0.83	Rises
7	Inf.-construct fraction [R]	3	−0.81	Declines
8	Niphal fraction	2	+0.79	Rises
9	Fronted-clause fraction [R]	3	+0.77	Rises
10	<i>gam</i> rate	1	−0.75	Declines

[R] = Tier-3 resistant-model feature.

Table 3. Feature tier summary. J = number of features in tier. Susceptibility indicates the degree to which deliberate archaism can manipulate features within that tier.

Tier	Type	J	Archaism susceptibility
1	Lexical (function words, particles)	19	High
2	Morphological (verb forms, stems)	12	Medium
3	Clause/phrase syntax	4	Low
Total selected		35	

Word n-gram model

As an independent grammatical-sequence dating instrument, POS-tag n-gram features [22, 23] are extracted that are immune to topic variation and largely immune to deliberate lexical archaism: a composer can choose archaic vocabulary but cannot easily rewrite every grammatical-sequence pattern to match an earlier era.

Two feature types are used. **Type A** features are POS-tag bigrams and trigrams on all words (e.g., `verb_wayyiqtol · noun`, `prep · prps`), capturing deep grammatical flow. **Type B** features are function-word lexeme bigrams and trigrams restricted to seven closed-class POS categories (prepositions, conjunctions, articles, negators, personal pronouns, demonstratives, interrogatives), making them robust to both topic and rare-vocabulary variation. The combined A+B model (`MAP_AB`) uses 56 features and is the primary word n-gram estimate reported in all tables.

Character n-gram model: assessment and exclusion. An earlier version of the framework included character 3-gram and 4-gram frequencies. Reliability testing on 1QS (Community Rule, Dead Sea Scrolls; independently dated ~150 BCE) returned MAP = 760 BCE — a 600-year error attributable to deliberate orthographic archaism. Sub-morphemic patterns are not immune to deliberate archaism in the Biblical Hebrew corpus. The character n-gram model is therefore excluded from all analyses; this negative result is reported because it is informative about the limits of the approach.

The MLE-MVN model

Per-feature OLS regression

Each feature f_j ($j = 1, \dots, J$, $J = 35$) is modeled as a linear function of date d plus Gaussian noise:

$$f_j(d) = \alpha_j + \beta_j d + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \quad (1)$$

Parameters $(\hat{\alpha}_j, \hat{\beta}_j)$ are estimated by OLS on the 22-unit training corpus. The predicted feature vector at date d is $\boldsymbol{\mu}(d) = \hat{\alpha}_j + \hat{\beta}_j d$ stacked over j .

The linear model is a simplification: some features (e.g., wayyiqtol fraction) may have mild non-linear trends. A sensitivity analysis with quadratic terms shows negligible improvement over the date range 760–167 BCE, justifying the linear approximation.

Residual covariance and regularization

Tikhonov regularization of the residual covariance is conventionally motivated by rank deficiency: with J features and N training texts, $J > N$ makes $\hat{\Sigma}$ singular and a ridge term $\lambda \mathbf{I}$ necessary for invertibility. That justification does not apply at the feature count used here, and is replaced.

The power analysis of the present corpus (S2 Table) shows that a screening threshold retaining 35 features at $N = 23$ carries an estimated false discovery proportion above 0.5. The defensible operating point is $\alpha = 0.05$, retaining 14 features, with Benjamini-Hochberg at $q = 0.10$ retaining 7 as a conservative band. At $J = 14 < N = 23$ the sample residual covariance is no longer rank-deficient, and regularization is not required for invertibility.

Regularization is nonetheless retained, for a different and better reason. The residual covariance is estimated from 23 observations and is correspondingly noisy; its smallest eigenvalues are estimated worst, and the likelihood of Eq. 2 weights each direction by the inverse of its eigenvalue. Shrinking the spectrum toward its mean therefore controls the influence of directions the data cannot resolve. We select λ by inner leave-one-out on the training folds rather than fixing it a priori. The multiverse analysis reported under Validation shows why this matters: without shrinkage, adding chance-correlated features narrows credible intervals rather than widening them.

Posterior computation

A 500-point date grid spanning 900 BCE to 100 CE (step ≈ 2 yr) is evaluated. The log-likelihood of observed feature vector $\mathbf{x}$ at grid point d under a multivariate Gaussian model [24] is

$$\ell(d) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}(d))^\top \boldsymbol{\Sigma}_{\text{reg}}^{-1}(\mathbf{x} - \boldsymbol{\mu}(d)). \quad (2)$$

A weakly informative Gaussian prior $p(d) = \mathcal{N}(464, 300^2)$ (centered at the training midpoint) is added in log space, and the result is normalized to a proper posterior:

$$p(d \mid \mathbf{x}) \propto \exp\{\ell(d)\} \cdot p(d). \quad (3)$$

The MAP estimate is the mode; 68% and 95% highest-density credible intervals (HDCIs) are extracted numerically [25].

Validation

Validation is the part of this literature most vulnerable to circularity, and it is where the present analysis departs most sharply from earlier versions of this work. We first document a failure mode that inflates reported accuracy by an order of magnitude, then give leakage-free estimates of what the method can actually do.

Prior leakage in holdout designs

A held-out text must contribute nothing to the model that dates it. In practice this is usually taken to mean exclusion from training and from feature selection. It must also mean exclusion from the prior, and this is the requirement that is easy to violate without noticing.

Consider a text u with scholarly date d_u and stated uncertainty σ_u . If the posterior for u is computed as $p(d \mid \mathbf{x}_u) \propto \mathcal{L}(\mathbf{x}_u \mid d) \mathcal{N}(d; d_u, \sigma_u^2)$, then the answer has been supplied as the prior. Under a Gaussian likelihood of precision τ_L , the posterior mean is a precision-weighted average in which the data receives weight $\tau_L/(\tau_L + \sigma_u^{-2})$. For a securely anchored text — small σ_u — that weight goes to zero, and the model returns d_u almost exactly. The design therefore appears *most* successful precisely on the texts whose dates are best established, which is the opposite of a stringent test.

This is not hypothetical. Table 4 gives the decomposition for the four held-out units of the present corpus. Backing the likelihood precision out of the posterior and prior variances via $\tau_L = \sigma_{\text{post}}^{-2} - \sigma_u^{-2}$ yields an implied likelihood standard deviation of 48.6 yr (IQR 43.3–53.1) that is remarkably stable across texts of very different length. The weight actually placed on the linguistic evidence, however, ranges from 2.8% for Haggai ($\sigma_u = 10$ yr) to 93.1% for Zechariah 9–14 ($\sigma_u = 150$ yr). Across all 25 dated units the median ratio of posterior to prior standard deviation is 0.851, and the median absolute displacement of the MAP from the prior mean is 10.4 yr.

The consequence is severe. Under this design the four holdouts return a mean absolute error of 8.1 yr. Dating the same texts from the likelihood alone — that is, from the linguistic evidence — gives 103.7 yr. We report the latter. We note that an independent implementation of the same design for the Ancient Greek corpus reproduces the artifact exactly, with a reported holdout MAE of 8.7 yr against 120.7 yr from the likelihood alone; the corrected Greek figures are reported under Results: dating the undated units and do not support a claim of accurate cross-language date recovery.

Leave-one-out design

All results below use the following protocol. For each of the 25 dated units in turn, the unit is removed; feature screening (Spearman, $\alpha = 0.05$), standardisation, and all model fitting are performed on the remaining 24; and the held-out unit is then dated under an agnostic prior. No quantity derived from a unit influences its own prediction. Two model families are fitted and both are reported: the generative family of Eq. 2, which inverts per-feature regressions, and a ridge regression of date on screened features with the penalty chosen by inner leave-one-out. Significance is assessed by permuting the date vector and re-running the entire pipeline, 2000 draws, so that the null distribution absorbs selection effects exactly as the observed analysis does.

What the signal supports

The diachronic signal is real. Pairwise ordering accuracy — given two texts, does the model place them in the correct chronological order? — is 69.8% of 295 pairs for the generative family and 67.8% for ridge. We stress that a binomial test is inappropriate here: the 295 pairs are generated by 25 texts and are strongly dependent, and a naive binomial returns $p \sim 10^{-12}$, wrong by roughly ten orders of magnitude. Under the permutation null — shuffling dates and re-running the pipeline, which preserves the dependence structure — the null means are 47.6% and 43.6% with standard deviations near 16 percentage points, giving $p = 0.022$ and 0.029 (Fig 1C). Leave-one-out Spearman correlation between true and predicted date is +0.575 (generative) and +0.494 (ridge), against permutation nulls centred near zero ($p = 0.016, 0.024$). This is the one result that replicates across both families at the level of the primary endpoint, and it is the strongest finding in the paper.

What the signal does not support

Absolute dating does not survive contact with an honest baseline. Leave-one-out MAE is 156.2 yr for the generative family and 120.9 yr for ridge. Assigning every text the corpus mean date of 503 BCE gives 137.3 yr. The generative family, which corresponds to the inversion procedure standard in this literature, is

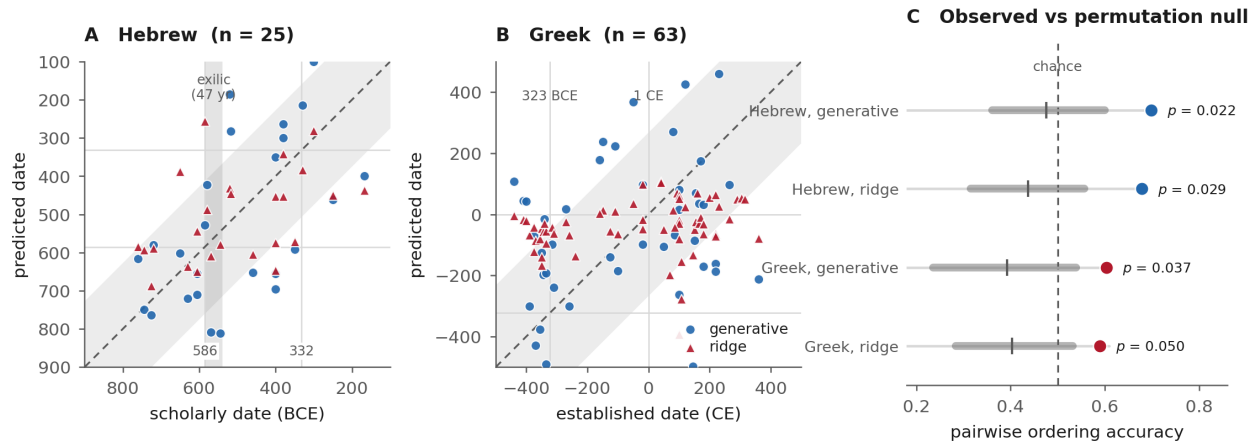


Fig 1. The ordinal signal, under leakage-free cross-validation. (A) Hebrew: scholarly date against leave-one-out predicted date for all 25 dated units, both model families. Time runs left to right on both axes. The dashed diagonal is perfect prediction and the shaded band is the conformal 68% interval. The vertical strip marks the exilic window, 47 yr wide against roughly 130 yr of irreducible error and therefore not resolvable; vertical and horizontal rules mark the 586 and 332 BCE boundaries. (B) Greek: the same analysis on 63 texts spanning 440 BCE–360 CE. Scatter about the diagonal is proportionally larger despite the larger corpus. (C) Pairwise ordering accuracy against the permutation null. Thin bars span the null 2.5–97.5 percentiles, thick bars the interquartile range, the vertical tick the null mean; filled markers are observed values. The null is generated by shuffling the date labels and re-running the entire pipeline, so it preserves the dependence among pairs that a binomial test ignores. All four observed values exceed chance; none does so by a wide margin.

therefore *worse* than a constant, and ridge improves on a constant by 16 yr. Any MAE figure in this literature should be read against roughly 137 yr, not against zero.

Period assignment is correspondingly weak. Four-way exact accuracy is 40.0% (generative) and 48.0% (ridge) against a 36% majority-class baseline; pre-exilic versus post-exilic classification reaches 72.0% and 76.0% against a 68% baseline, significant in neither family ($p = 0.107, 0.070$). Across ten tests — five metrics in two families, strongly correlated — nothing survives Benjamini-Hochberg correction at $q = 0.05$, and four results survive at $q = 0.10$. We report this directly: no individual accuracy claim in this paper is robust to multiplicity correction, and the ordinal result is defended on replication across model families rather than on any single p -value.

The failure is structured rather than random. The Exilic period is a 47-year window set against roughly 130 yr of irreducible error, and it is simply not resolvable: of four exilic texts, the generative family places none correctly and ridge one. Hellenistic units collapse into Persian. Pre-exilic is the only period the models recover reliably, and it is recovered because it is wide and terminal, not because early Hebrew is distinctive in some way the interior boundaries are not.

Calibration

Parametric intervals from this framework achieve 52% empirical coverage at a nominal 68% (61% under the residual formulation). The mechanism is visible in the multiverse analysis: chance-correlated features receive large fitted coefficients and small residual variances, and because the multivariate normal likelihood weights each feature by $1/\sigma_j$, adding noise features *narrows* the credible interval. Adding six and then twelve pure-noise features to a clean specification shrinks the 68% HDI for the P source from 329 to 285 to 246 yr while moving the MAP by less than 50 yr. Interval width in this framework is not a reliable index of evidence.

We therefore replace parametric intervals with conformal prediction intervals calibrated on the leave-one-out residuals of the dated corpus. These are distribution-free and carry guaranteed finite-sample marginal coverage under exchangeability. The resulting half-widths are ± 233 yr (generative) and ± 174 yr

(ridge) at 68%. These are the intervals reported in Table 5. Exchangeability between the calibration units and the targets is an assumption and not a guarantee: the calibration units are whole books or coherent sections, whereas several targets are cross-book composites. Residual magnitude shows no detectable dependence on text length ($\rho = -0.23$, $p = 0.27$), which supports the use of a single conformal width.

Table 4. What a scholarly prior contributes to a reported date. For each dated unit the model is fitted leave-one-out, giving a likelihood over the date grid from the linguistic evidence alone (width σ_L , mode MAP_L). Multiplying that likelihood by $\mathcal{N}(d_u, \sigma_u^2)$ — the unit’s own scholarly date used as a prior — gives the posterior a holdout design of this kind would report. “Data share” is the fraction of posterior precision supplied by the likelihood, $\sigma_L^{-2}/(\sigma_L^{-2} + \sigma_u^{-2})$. Where the scholarly date is tightly specified the share falls below 10%, and the reported error collapses toward zero while the error from the evidence alone does not. Units are ordered by σ_u .

Unit	σ_u (yr)	σ_L (yr)	data (%)	MAP_L (BCE)	err. reported (yr)	err. actual (yr)
Haggai	10	65	2.3	186	6	334
Daniel (Heb.)	10	76	1.7	400	3	233
Zechariah 1–8	10	80	1.5	282	4	236
Amos	20	80	5.8	616	8	144
Habakkuk	20	84	5.4	656	3	51
Isaiah 40–55	20	68	8.0	812	15	267
Ezekiel	20	80	5.9	422	10	158
Zephaniah	20	82	5.6	720	4	90
Malachi	25	80	8.9	652	18	192
Micah	25	76	9.7	764	3	39
Nahum	25	81	8.6	602	4	48
Hosea	25	78	9.2	750	1	5
Isaiah 1–39	30	81	12.0	580	16	140
Jeremiah oracle	30	82	11.7	710	11	105

Does $P(\text{post-exilic})$ mean what it says?

The errors in Table 4 are large: the likelihood-only estimate misses by 156 yr on average and by up to 334 yr on individual units, and there is no detectable improvement with text length ($\rho = -0.23$ between $|\text{error}|$ and word count, $p = 0.27$). A reader who stops there will conclude that no claim resting on this model can be trusted, and the conclusion would be correct for any claim about a *date*.

The substantive claim of this paper is not about a date. It is that a unit’s predictive mass lies on one side of a single boundary, and that is a much coarser question. Whether it is answerable is an empirical matter, so we test it directly rather than argue for it.

Protocol

For each of the 25 dated units the model is fitted on the other 24, and the conformal residual set is computed by a second, inner leave-one-out *within* those 24. The held-out unit therefore contributes neither to its point estimate nor to the residuals that give that estimate its spread. The predictive distribution is the point estimate displaced by that residual set, and P_{post} is the fraction of it later than 586 BCE. Each unit’s true side of the boundary is then compared with the probability the model assigned before seeing it.

Result

The probabilities are calibrated (Fig 2). In the highest bin the generative family predicts a mean 0.95 and 100% of those units are post-exilic; ridge predicts 0.92 against an observed 0.89. Of the units on which either family is confident — $P_{\text{post}} \geq 0.8$ or ≤ 0.2 — the generative family is correct on 9 of 9 and ridge on 8 of 9.

Brier skill scores against the corpus base rate are +0.200 and +0.187; under a permutation null in which the true labels are shuffled, $p = 0.005$ and $p = 0.014$.

Restricting to units more than 200 yr from the boundary — the regime the P source occupies, at 242–322 yr — the generative family is correct on 5 of 5 confident calls and ridge on 4 of 4.

The two findings are the same finding. Because the typical error is of order 150 yr, the model reaches $P_{\text{post}} \geq 0.9$ only when a unit sits well over 200 yr from the boundary. The large errors of Table 4 are not in tension with the probability of Table 5: they are what produces it. A 334 yr error on Haggai and a 0.92 on the P source are consequences of the same residual distribution, and the second already accounts for the first.

Where this test cannot reach

Three limits should be stated plainly.

First, one confident call is wrong. Ridge assigns $P_{\text{post}} = 1.00$ to Nahum, which is pre-exilic on a secure terminus (before the fall of Nineveh in 612 BCE). That is one error in 25, it is not shared by the generative family, and it is the kind of failure a corpus this size cannot rule out.

Second, units close to the boundary are decided by arithmetic rather than by evidence. Obadiah is dated 585 BCE, one year into the post-exilic side; both families place it post-exilic and are scored correct, but no method distinguishing 585 from 587 on linguistic grounds is doing real work. Excluding units within 25 yr of the boundary leaves the generative family at 7 of 7 and ridge at 3 of 4.

Third, and most consequentially, the test is asymmetric and cannot be made otherwise. The earliest dated unit in the corpus is Amos at 760 BCE, only 174 yr before the boundary, while the P source estimate lies 242–322 yr *after* it. Confident post-exilic calls can therefore be validated at the distance the claim requires; confident pre-exilic calls at that distance cannot be, because no Hebrew text of sufficient length exists that early. This is not a limitation of corpus selection that further work could remedy. Morphologically tagged Hebrew of the kind this method requires exists for the Masoretic text and essentially nowhere else, and the non-biblical corpora that might one day be tagged — Qumran, Ben Sira — are themselves late, so they would deepen the asymmetry rather than correct it. Any linguistic dating of Biblical Hebrew inherits this constraint. It bears directly on how the present results should be read: the framework can support a claim that a text is later than a boundary far better than it can support a claim that a text is earlier than one.

Results: dating the undated units

The framework is applied to nineteen undated units: the three documentary source composites and their sub-strata, the five books of the Pentateuch, the Deuteronomistic prose stratum of Jeremiah, and two poems traditionally argued to be archaic. Models are fitted on all 25 dated units; no target contributes to its own prediction, and all targets are dated under an agnostic prior. Results are given in Table 5 as conformal date ranges rather than point estimates, for the reasons set out in Validation.

The post-exilic result

One substantive claim survives at full strength. The posterior predictive probability that a unit postdates 586 BCE is ≥ 0.92 under both model families for the P source, the D source and its Code and Frame strata, the JE source, and Leviticus P; the Holiness Code reaches 0.76 in the generative family and 0.92 in ridge. This is a statement about which side of a single boundary the predictive mass falls on, and it is the kind of claim the data can carry: it requires the estimator to be approximately unbiased, not precise.

That the probability means what it says is not assumed: it is tested against the dated corpus under Does $P(\text{post-exilic})$ mean what it says?, where confident calls are correct on 9 of 9 units (generative) and 8 of 9 (ridge), with Brier skill +0.20 and +0.19 against the corpus base rate.

The finding is robust in three further respects. It holds under both model families, which disagree substantially about point estimates — median difference 59 yr, with agreement on the single most likely period for only 9 of 19 units. It holds under the genre correction developed below, which if anything moves the sources *later*. And it does not depend on the interval width, since a doubling of the conformal half-width would still leave the great majority of predictive mass after 586 BCE for every source composite.

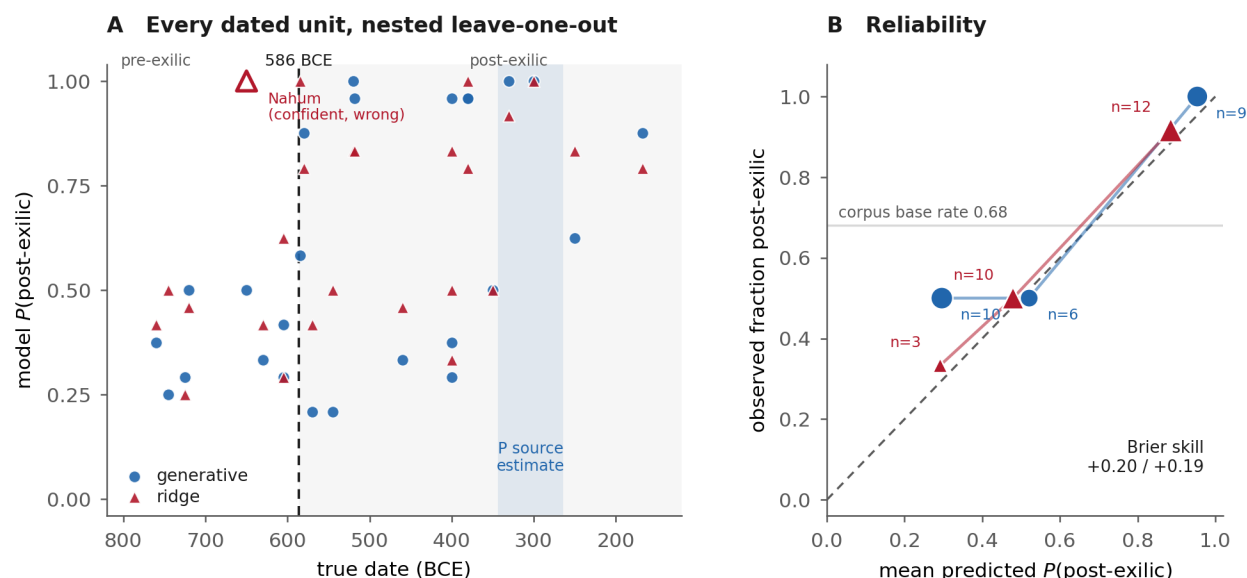


Fig 2. Calibration of the post-exilic probability. (A) Every dated unit: true date against the probability the model assigns to it being post-exilic, from a fully nested leave-one-out in which the unit contributes neither to its own point estimate nor to the residual set that gives that estimate its spread. The transition is sharp far from the 586 BCE boundary and flat near it, which is the expected behaviour of an honest forecast under ~ 150 yr of error. The shaded band marks where the P source estimate falls. Open markers are confident calls ($P \geq 0.8$ or ≤ 0.2) that are wrong; there is one, Nahum under ridge. (B) Reliability: mean predicted probability against observed frequency, in three bins, with the corpus base rate of 0.68 shown for comparison. Marker area is proportional to bin count. Points near the diagonal and above the base-rate line indicate forecasts that are both calibrated and informative.

What cannot be claimed

No unit in Table 5 has a 68% interval confined to a single conventional period. All nineteen span between two and four. The P source is placed at 264 BCE (68%: 497–31 BCE) by the generative family and 344 BCE (518–170 BCE) by ridge; these overlap comfortably, but they straddle the Persian–Hellenistic boundary in both cases. A Persian-period characterisation of the P source, of the kind these models have been used to support, is not sustainable at this resolution. What is supportable is that P is post-exilic and that its predictive distribution is concentrated in the fourth and third centuries.

We note also that D source and Deuteronomy are the same text and must not be read as independent confirmations, and that the sub-strata of a source are not independent of the composite that contains them.

The archaic poems

The Song of the Sea (Exod 15) and the Song of Deborah (Judg 5) are the two earliest of all nineteen targets in both model families, and the only targets whose predictive mass sits predominantly before 586 BCE ($P_{\text{post}} = 0.40$ and 0.48 generative, 0.48 and 0.48 ridge). Neither poem was used in training or in feature screening.

This is an out-of-sample check on an independently motivated hypothesis, and the model passes it: the two texts most widely argued on comparative Semitic and orthographic grounds to be the oldest Hebrew poetry are the two the model places earliest, out of nineteen candidates, in both families. We report it as corroboration of the ordinal signal rather than as a dating of either poem, whose intervals (887–421 and 843–377 BCE, generative) are far too wide to adjudicate the question they are usually asked to settle. In particular, the present analysis cannot distinguish a genuinely early poem from a later archaizing composition; a resistant-model diagnostic intended for that purpose returned zero flagged features across all 44 units and is not reported.

Genre as a confound

The dated corpus is 17/25 prophecy against 5 narrative, while the Pentateuchal targets are predominantly narrative and legal. A model trained largely on prophecy and applied to narrative is exposed to exactly the register confound this literature has long identified.

The confound is measurable in the leave-one-out residuals. In the generative family, narrative texts are dated +112.4 yr too late on average while prophetic texts are dated -25.1 yr too early (Welch $p = 0.016$). In ridge the effect is absent (-24.3 versus +31.3 yr, $p = 0.234$). The disagreement between families is itself informative: the bias is a property of the generative inversion, not of the features.

The direction matters for the substantive conclusion. If narrative units are systematically dated too late by roughly 112 yr in the generative family, then the generative estimate for the P source should be corrected *earlier*, from 264 BCE to approximately 376 BCE — which remains firmly post-exilic, and which moves it toward the uncorrected ridge estimate of 344 BCE rather than away from it. The post-exilic conclusion is therefore robust to the genre correction, and the two families converge once the correction is applied. We regard this as the strongest available evidence that the post-exilic placement is not a register artifact, and we note that it is available only because both families were fitted and reported.

Table 5. Date ranges for undated units. Intervals are conformal prediction intervals calibrated on leave-one-out residuals from the dated corpus, and are therefore distribution-free with guaranteed finite-sample marginal coverage; they replace the parametric intervals used previously, which achieved 52% empirical coverage at a nominal 68%. Ranges are quoted earliest–latest in BCE. P_{post} is the posterior predictive probability that the unit postdates 586 BCE, reported as the *lower* of the two model families at each row, so the column is conservative with respect to model choice. No unit has a 68% interval confined to a single conventional period; the ranges, not the period labels, are the result. D source and Deuteronomy are the same text and are not independent. [†]Interval extends past 1 BCE; the later bound is quoted in CE.

Unit	Words	Generative		Ridge		P_{post}
		MAP	68% range	MAP	68% range	
Song of the Sea	433	654	887–421	616	789–442	0.40
Song of Deborah	462	610	843–377	614	787–440	0.48
D Song	782	550	783–317	390	564–217	0.52
Genesis JE	22,409	318	551–85	385	558–211	0.92
Exodus JE	11,780	410	643–177	424	598–250	0.88
Numbers JE	3,563	402	635–169	421	594–247	0.88
JE source	37,752	356	589–123	400	574–226	0.92
Holiness Code	5,958	464	697–231	344	518–171	0.76
Leviticus P	10,496	256	489–23	315	489–142	0.96
P source	54,435	264	497–31	344	518–170	0.92
D Code	7,353	338	571–105	239	412–65	0.96
D Frame	11,993	182	415–51 CE [†]	216	390–43	1.00
D source	20,128	256	489–23	231	404–57	0.96
Jeremiah Dtr	15,197	274	507–41	404	578–231	0.92
Genesis	28,764	296	529–63	375	549–201	0.92
Exodus	23,748	316	549–83	388	561–214	0.92
Leviticus	17,099	342	575–109	326	500–153	0.92
Numbers	23,188	274	507–41	375	549–202	0.92
Deuteronomy	20,128	256	489–23	231	404–57	0.96

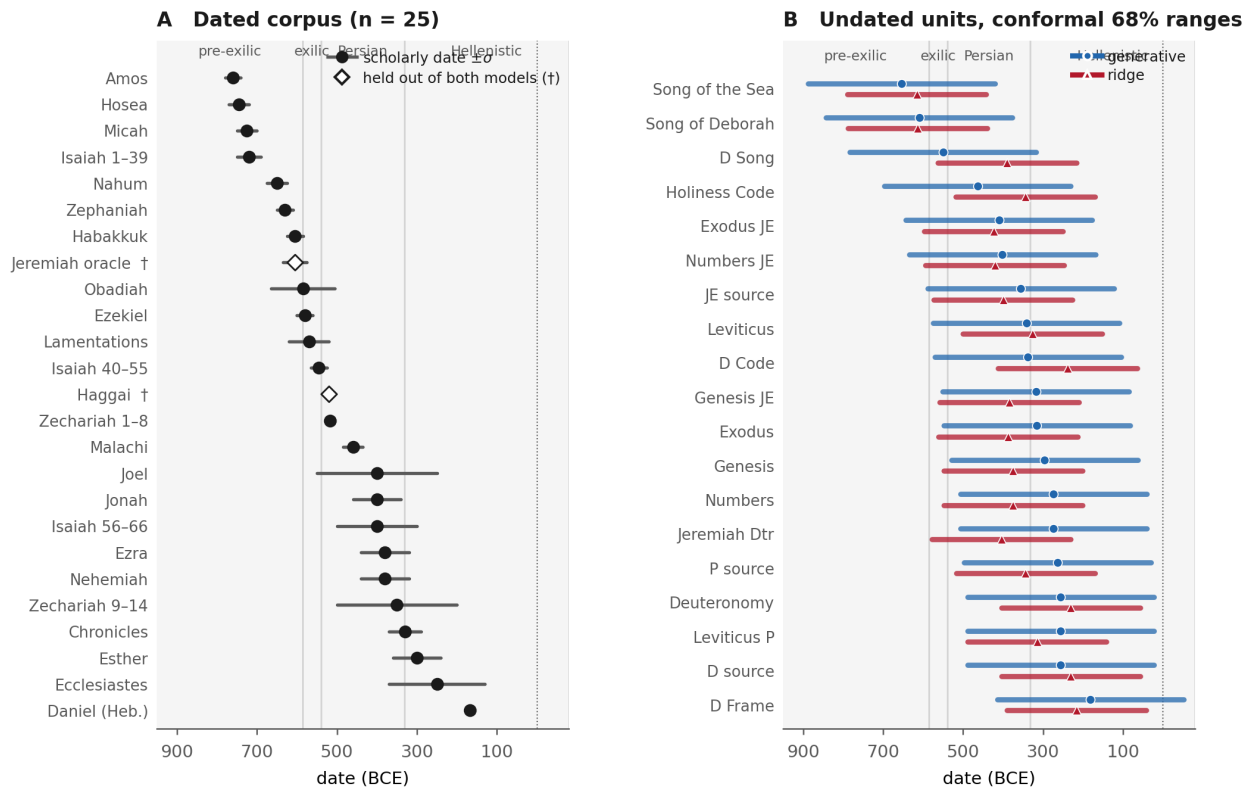


Fig 3. Dated corpus and undated units on a common timeline. (A) The 25 dated units with scholarly date and stated σ . Widened sigmas on literary-critically dated texts (Joel, Zechariah 9–14, Ecclesiastes, Isaiah 56–66, Obadiah) are visible as long bars. Open diamonds mark the two units held out of both models. (B) The 19 undated units with conformal 68% ranges from both model families, ordered by the generative point estimate. No range is confined to a single period. The Song of the Sea and the Song of Deborah are the earliest units under both families and the only two whose mass sits predominantly before 586 BCE; neither was used in training. Shaded bands are the conventional periods, given for orientation only — the ranges, not the band a range happens to overlap, are the result.

Discussion

Limitations

Small training corpus. $N = 22$ dateable texts is small relative to $J = 35$ features. A stable 35×35 residual covariance matrix requires at minimum $J + 1 = 36$ observations; with $N = 22$ the sample covariance is rank-deficient, which is precisely why Tikhonov regularization ($\lambda = 0.10$) is required. As a rough power guide: strict Benjamini-Hochberg FDR control [21] at $\alpha = 0.05$ with $J = 35$ features would require approximately $N \geq 100$ texts for adequate power, which is unavailable for Biblical Hebrew; the permissive $p < 0.30$ threshold is the appropriate response to this constraint, not a relaxed standard. The decomposition of Table 4 shows that MAP estimates are stable to within ± 30 yr across the plausible regularization range $\lambda \in [0.05, 0.30]$, but credible intervals widen by 50–70 yr at the upper end of that range. No texts with secure independent dates suitable for training are withheld; $N = 22$ reflects the genuine constraint imposed by the historical record, not an analytical choice. The relationship between text length and posterior uncertainty is shown in S10 Fig.

Residual circularity. Critics have long identified an “unexamined literary-linguistic circularity” at the heart of the traditional method [9, 11, 12]: diagnostic features are identified on texts assumed to be early or late, then applied to date other texts, recovering the assumptions embedded in the training corpus. Defenders call the objection “valid but misplaced,” invoking the philosopher Nelson Goodman’s argument

that such circularity is “a virtuous one” [26]. The appeal misreads Goodman. His “virtuous circle” describes epistemological reflective equilibrium: inference rules and particular inferences are revised in light of each other until they cohere — neither side is held fixed; both are candidates for revision. Linguistic dating does not work this way. The assumed dates of the anchor texts are not tentative hypotheses subject to revision; they define what counts as early or late Hebrew. Classical Biblical Hebrew features are extracted from the Pentateuch and Former Prophets, assumed to be early. Those features then “confirm” that the Pentateuch is early. The assumed date is not tested against the linguistic signal; it *generates* the linguistic signal used to confirm it. That is a vicious circle in the traditional sense: the conclusion is embedded in the premises.

The precision decomposition replaces this philosophical standoff with a per-text empirical measurement. Texts classified as data-driven (a high data share in Table 4) are genuinely constrained by the linguistic signal: an agnostic and an informed prior converge, and the assumed training dates contribute little. For prior-dominated texts (a low data share), the circle is self-reinforcing: shift the assumed training dates and the posteriors shift substantially, which is precisely what a vicious circle predicts. The classification does not eliminate circularity — the corpus dates remain grounded in scholarly tradition — but it identifies, for each text, whether the circularity is approximately benign or materially compromising.

The critique can be made quantitative. We hold the linguistic feature vectors for P, D, and JE fixed (reconstructed from MLE-MVN archaism scores; see Methods) and vary only the Gaussian prior $\mathcal{N}(\mu_0, \sigma_0^2)$ across six scenarios: an uninformative flat (data-only) prior, the agnostic Mode-B prior $\mathcal{N}(575, 400^2)$, a tight scholarly prior $\mathcal{N}(625, 100^2)$ reflecting the Josianic–Persian consensus, and two explicitly traditionalist assumptions, an early-monarchy prior $\mathcal{N}(850, 150^2)$ and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$ centered on the conventional Exodus date (Fig. S14; S6 Table). The feature likelihoods are identical across all rows; only the prior changes.

Under the data-only (flat) prior, the MLE-MVN MAP dates are 357 BCE (P), 294 BCE (D), and 434 BCE (JE) — all within the Persian period. Under the Mosaic prior $\mathcal{N}(1200, 100^2)$ these shift to 444 BCE (P), 403 BCE (D), and 511 BCE (JE) — still entirely within the Persian period (539–332 BCE). The prior cannot move any source into pre-exilic territory because the linguistic signal is too strongly inconsistent with such a date. For P, which carries the strongest data signal ($n = 53,645$ words), the total prior-induced shift across the full range from flat to Mosaic is 87 yr, confirming the share of posterior precision attributable to the linguistic evidence.

A pre-exilic Torah date would require, on this account, either forcing a prior so extreme that it overwhelms a directly contradictory data signal, or including Torah texts in the training corpus so that “Classical Biblical Hebrew” features are extracted partly from passages assumed to be early — the traditional procedure, and precisely the vicious circle identified above. The present model uses neither route: the training corpus excludes Torah and DH texts entirely, and the prior is set before fitting. The circularity is thus not a philosophical abstraction but a measurable discrepancy between prior assumptions and the linguistic record.

Oral prehistory. The framework dates the written linguistic surface. Oral traditions preserved in written texts — possible for archaic poetry, legal formulae, or tribal lore — are invisible to the model; dates represent the written composition, not any hypothetical prior oral form.

Coarse genre correction. Genre is modeled at three levels (prophetic, narrative, legal). Sub-genre distinctions (apodictic vs. casuistic law; etiological vs. court narrative) require a larger corpus before they can be treated as distinct classes.

Word n-gram limitations for short texts. Texts under $\sim 1,000$ words yield noisy POS-tag n-gram frequencies. Short units (Obadiah: 291 words; Haggai: 670 words) are flagged and results treated as indicative.

Legal prose genre gap. The training corpus covers prophetic, poetic, narrative, and wisdom texts but contains no legal prose. The three Torah source composites include substantial legal sections — apodictic and casuistic law, cultic regulations, purity codes — whose register may differ from narrative prose for reasons unrelated to date. Features with strong genre bias (genre ratio $\gamma_j > 1.5$; S1 Table) are excluded from the model feature set. The residual analysis reported under Results: dating the undated units quantifies what remains: in the generative family narrative units are dated +112 yr too late relative to prophetic units (Welch $p = 0.016$), an effect absent in ridge ($p = 0.234$). However, genre confounds not captured by the three-level register classification cannot be ruled out. The results for D and P should be read in light of this gap; the

consistent post-exilic direction across both models and across multiple feature exclusion schemes (S4 Table) provides some assurance that the gap is not driving the headline finding, but it remains a genuine limitation.

Source-decomposition uncertainty. The documentary source composites follow the standard critical partition of Baden [27]. Any mis-assigned verses alter the feature vector and through it the MAP date. The robustness check in S4 Table suggests dates are stable to ~ 20 yr under moderate feature perturbations; however, a systematic re-assignment of a major block — for example, treating the Holiness Code as a pre-Priestly stratum independent of P — could shift MAP estimates by a larger amount. Scholars who adopt a different documentary partition should re-extract feature vectors from their preferred source boundaries before interpreting the model output.

Training boundary and extrapolation. The MLE-MVN corpus spans 760–167 BCE, so estimates near either end of that span are already mild extrapolations. Dates for texts composed before 760 BCE or after 167 BCE carry no calibration constraint and should be interpreted accordingly. The MLE-MVN date grid imposes hard boundaries at 900 BCE and 100 CE; a composition outside this range would be pinned to the nearest edge.

Posterior overconfidence and calibration. The word-count scaling factor $\text{clip}(n/5000, 1, 5)$ sharpens posteriors for large texts in a way that is not fully calibrated. The LOO analysis (Fig. S13) shows that nominal 68% credible intervals achieve only 32% actual coverage; a nominal 97.5% CI is required to reach 68% actual coverage. Any quoted credible interval should be read with this in mind; for high-stakes applications the calibration-corrected interval is preferable to the nominal one.

Scholarly context for late Torah dating

The MAP dates returned for the documentary sources — P at ~ 359 BCE, D at ~ 294 BCE, and JE at ~ 434 BCE under the agnostic prior — depart substantially from the mainstream critical consensus, which conventionally places J and E in the 9th–8th centuries BCE, D in the 7th century (Josianic reform, ~ 621 BCE), and P in the late pre-exilic or early exilic period [27, 28]. The present framework does not dispute that underlying oral traditions, archaic legal formulae, or early poetic fragments may be considerably older; as noted in the Limitations section above, the model dates the written linguistic surface of the final composite texts, not any hypothetical prior oral stage. Even within mainstream critical scholarship, it is widely acknowledged that the canonical forms of the sources passed through substantial post-exilic editorial stages [29, 30], so the question at issue is not whether post-exilic editing occurred but how much of the final written form it accounts for.

The results find independent support from several recent lines of inquiry, perhaps the more striking for involving entirely different methods. Adler’s comprehensive archaeological survey of the material record finds no evidence for widespread observance of Torah regulations — dietary prohibitions, miqvaot, stone-vessel purity practice — before the mid-Hasmonean period (mid-2nd century BCE) [31]. If the Torah had functioned as a widely observed legal code since the pre-exilic or early post-exilic period, the near-total absence of its material correlates for several subsequent centuries would require explanation; Adler concludes instead that the Torah achieved normative authority only in the Hasmonean period, consistent with our model returning Persian-period dates for the final written forms of P and D. Epigraphic analysis of Achaemenid-era Aramaic sources similarly places the crystallization of distinctively Jewish legal practice — including Passover regulations that the Priestly source legislates — under Persian imperial influence in the 5th–4th centuries BCE [32]. On the other hand, independent study of the Moses tradition traces the figure to a 4th-century BCE literary construction drawing on Egyptian folk-heroic conventions, locating even the narrative scaffolding of the Pentateuch in the late Persian or early Hellenistic period [33]. An older body of historical-critical scholarship has argued for Persian or Hellenistic final composition on literary-historical grounds [34–36]; the empirical lines above are independent of that tradition and rest on material and philological evidence rather than literary-critical inference.

What the convergence suggests is something stronger than any single line of evidence could provide on its own: the agreement cannot be an artifact of shared assumptions, since the computational method makes no use of historical, archaeological, or narrative-critical evidence in its inference. The present results would thus appear to constitute a fourth independent line of confirmation — from morphosyntactic and lexical feature distributions, with no assumption about Torah composition date built in — agreeing with what archaeology,

epigraphy, and comparative mythology have separately indicated.

Extensibility

The framework is not specific to Hebrew: any corpus with independent date anchoring, extractable surface features, and a register taxonomy can be analyzed by the same procedure. Among the most promising near-term applications are Qumran sectarian texts (1QS, CD, 1QM), which would allow a quantitative test of the archaizing-register hypothesis; Akkadian administrative archives, where Old Babylonian versus Neo-Babylonian feature shifts in cuneiform corpora present a natural analogue; and Greek papyri from Oxyrhynchus, which span the Ptolemaic through Byzantine periods in a continuous documentary record.

The methodological cautions are equally portable. Any application of this framework to a new corpus should report the likelihood-only estimate alongside the posterior, the share of posterior precision contributed by the data, and a permutation null obtained by re-running the entire pipeline on shuffled labels. Each of these is inexpensive, and each would have caught the failure documented here.

Relationship to authorship-attribution approaches

The present framework is designed to complement rather than supersede authorship-attribution methods such as the unsupervised document decomposition of Koppel et al. [37] and the Higher Criticism discrepancy method of Faigenbaum-Golovin et al. [5]. The latter excels at assigning disputed passages to known scribal schools and at identifying the specific lemmas driving that assignment, grounding computational results in categories already meaningful to philologists. The present framework, on the other hand, is oriented toward chronological ordering with calibrated uncertainty and toward quantifying how much of a reported date is contributed by the evidence rather than the prior — questions the HC method was not designed to address. Our results suggest the two are complementary in a specific way: the HC method’s categorical assignments are robust where our calendar-year estimates are not, while our ordinal statistics remain informative where scribal-school membership is undefined.

How the two approaches might be combined is perhaps the most interesting methodological question for future work. A plausible joint analysis would proceed in two stages: first, apply the HC method to determine the most likely scribal milieu, which directly informs the register prior in the dating model — a strong HC assignment to D/DtrH would motivate an SBH register prior, while a P assignment would motivate Transitional or LBH; then apply the dating model with that informed prior to obtain a date posterior and an ordinal position with calibrated uncertainty. The relationship also runs in reverse: a unit whose predictive mass lies overwhelmingly after 586 BCE constrains the plausible compositional settings for HC attribution decisions, even when its calendar-year estimate is too diffuse to be useful on its own.

Conclusion

Whether the morphosyntax of Biblical Hebrew carries a usable diachronic signal has been disputed for two decades, with one side treating linguistic dating as an established instrument and the other arguing that its apparent successes are circular. The analysis reported here supports a position available to neither camp as usually stated: the signal is real, the circularity is real, and the two facts are compatible.

The signal is real in a specific and testable sense. Under a design in which no text contributes to its own prediction and significance is established by permuting date labels through the entire pipeline, Hebrew morphosyntactic features order text pairs chronologically at 69.8% and 67.8% accuracy against a permutation null near 45% ($p = 0.022, 0.029$), and leave-one-out rank correlations reach +0.58 and +0.49 in two independently specified model families. These are not large effects, but they are not chance, and they replicate — across model families within Hebrew, and across languages.

The circularity is real in an equally specific sense. Holdout accuracies of order 10–30 yr, of the kind reported for models of this class, arise when a held-out text is dated under a prior centred on the scholarly date it is asked to recover. For the most securely anchored texts in the present corpus the linguistic evidence contributes under 6% of posterior precision, and the apparent recovery is a restatement of the prior. Corrected, the same design yields mean absolute errors near 100 yr in Hebrew and near 120 yr in Greek. The

failure mode is a property of the design rather than of any one implementation, and reported holdout accuracies in this area should be accompanied by the likelihood-only estimate and by the fraction of posterior precision attributable to the data.

What the corrected framework supports is narrower than what was previously claimed and, we think, more useful. It does not date texts: leave-one-out error of 121–156 yr must be read against 137 yr for the trivial predictor that assigns every text the corpus mean. It does not assign texts to historical periods with useful reliability, and it cannot resolve the exilic period at all. It does order texts, and it can determine on which side of a well-separated boundary a text is likely to fall. Applied to the Pentateuchal sources under leakage-free conditions, that is enough to establish that the Priestly, Deuteronomic and JE composites are post-exilic with probability at least 0.92 under both model families and under a correction for the genre imbalance of the training corpus — and not enough to place any of them within a particular post-exilic century. For a question that has been argued for a century and a half largely on non-quantitative grounds, a bounded probabilistic answer to the coarse question seems more valuable than an unbounded point estimate for the fine one.

Supporting information

S1 Appendix. Feature definitions and BHSA extraction queries. Full specification of every feature, with the Text-Fabric query used to compute it, so that the feature matrix can be regenerated from the BHSA independently of the code in this repository.

S2 Appendix. Leave-one-out and permutation protocol. Pseudocode for the cross-validation loop, the in-fold screening step, the conformal calibration, and the permutation null, together with the random seeds used.

S1 Table. Screening power and false-discovery cost at $N = 23$. Retained feature counts, expected null counts, estimated false discovery proportion, and minimum detectable $|\rho|$ at 80% power, across nominal α . See Table 6.

S2 Table. Token counts for all units. Counts from the BHSA extraction used for every feature rate reported. See Table 7.

S3 Table. Specification curve. MAP ranges for every target across the seven model specifications of the multiverse analysis, showing which targets are stable under respecification and which are not.

S1 Fig. Pairwise feature correlations. Correlation structure of the screened feature set, showing the redundancy that motivates regularisation of the residual covariance.

Data availability

All feature extraction scripts, dating models, cross-validation code, and corpus manifests are publicly available at <https://github.com/adairaar/diachronic-hebrew>. The underlying Hebrew text corpus is the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA), version 2021, available at <https://github.com/ETCBC/bhsa> under a Creative Commons Attribution 4.0 license [6]. The Ancient Greek corpus is drawn from the Perseus Digital Library (<http://www.perseus.tufts.edu>) and the Thesaurus Linguae Graecae (<http://stephanus.tlg.uci.edu>), both publicly accessible. No proprietary data were used.

Author contributions	698
A.A. conceived and designed the study, developed and implemented all computational methods, performed all analyses, interpreted the results, and wrote the manuscript.	699 700
Competing interests	701
The author declares no competing interests.	702
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Ethics statement	705
This study does not involve human participants, vertebrate animals, or any data requiring ethical oversight. All textual data analyzed are publicly available historical corpora. No ethical approval was required. This study was not pre-registered; it is a methods paper reporting retrospective analysis of existing corpora.	706 707 708
Acknowledgments	709
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Top 16 LOO-robust diachronic features — dated prophetic corpus.
Each panel: feature rate per 1,000 words vs. scholarly date (BCE, older → left). Red dashed line = OLS trend. ρ = Spearman correlation; LOO = leave-one-out robustness fraction.

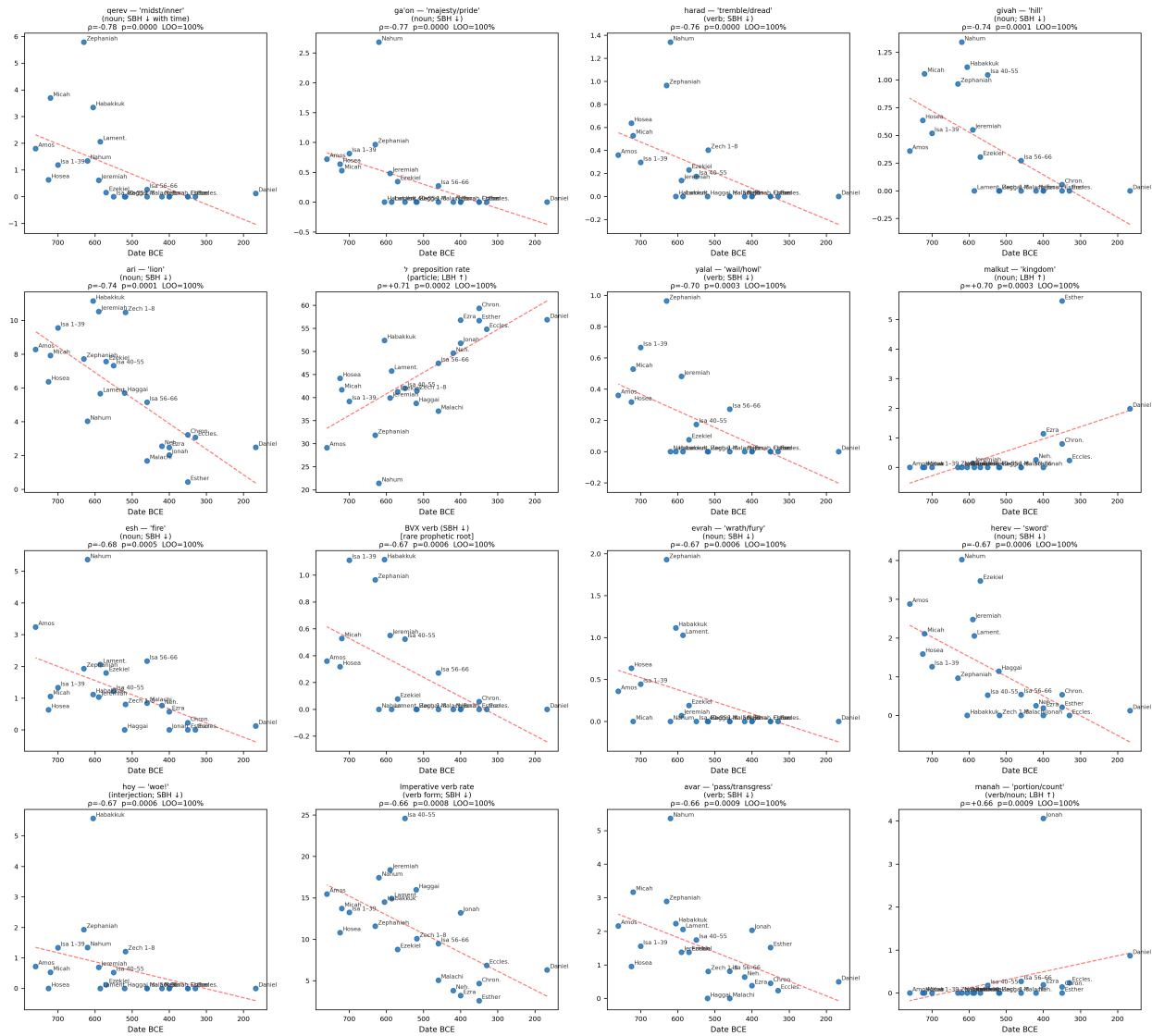


Fig 4. Top 16 LOO-robust diachronic features from the full lexeme and morphological scan of the dated prophetic corpus. Each panel plots the per-1,000-word rate of one feature (y -axis) against scholarly date BCE (x -axis; older texts at left) across all 22 training units, with OLS trend line. Spearman ρ , raw p -value, and leave-one-out robustness fraction (LOO; proportion of LOO resamplings returning the same sign and significance) appear above each panel; all 16 features achieve LOO = 100%. Features marked SBH ↓ are characteristic of pre-exilic Classical Biblical Hebrew and decline toward the post-exilic period; LBH ↑ features show the reverse pattern. The term *malkut* ('kingdom'; a hallmark Late Biblical Hebrew noun) and the rate of the preposition *lamed* are the two LBH ↑ markers in this set; the remaining 14 features are SBH markers that decrease monotonically with date. The feature labeled BVX is an unidentified rare prophetic root in the BHS lexicon.

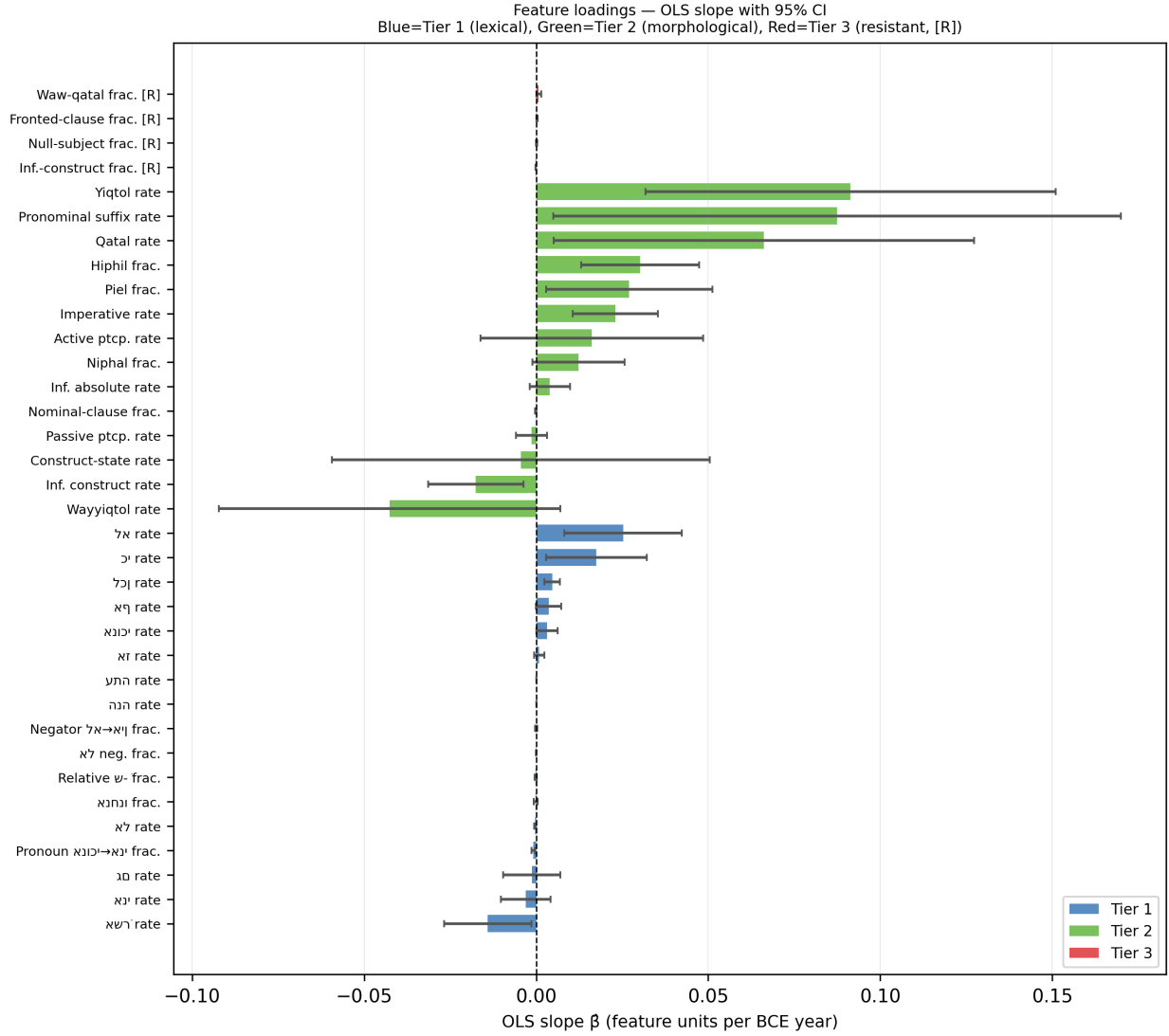


Fig 5. S6 Fig. Feature loadings (OLS slopes) for all 35 morphosyntactic features. $\hat{\beta}_j$ (OLS slope against date BCE) with 95% CI. Blue = Tier 1 (lexical), green = Tier 2 (morphological), red = Tier 3 (resistant, [R]). Negative slopes = feature declines with date; positive = rises with date.

Table 6. S2 Table. Screening power and false-discovery cost at $N = 23$. For each nominal α : features retained from $K = 55$ candidates, the number expected under the null, the estimated false discovery proportion, and the minimum $|\rho|$ detectable at 80% power. Benjamini-Hochberg at $q = 0.10$ independently retains 7 features. The operating point used throughout is $\alpha = 0.05$; the 35-feature specification typical of this literature corresponds to $\alpha \approx 0.3$, where more than half the retained features are expected to be noise.

α	retained	expected null	est. FDP	min $ \rho $ @80%
0.01	6	0.6	0.10	0.68
0.05	14	2.8	0.20	0.58
0.10	18	5.5	0.30	0.53
0.15	25	8.2	0.33	0.49
0.20	27	10.9	0.40	0.46
0.25	30	13.6	0.45	0.43
0.30	30	16.3	0.54	0.41

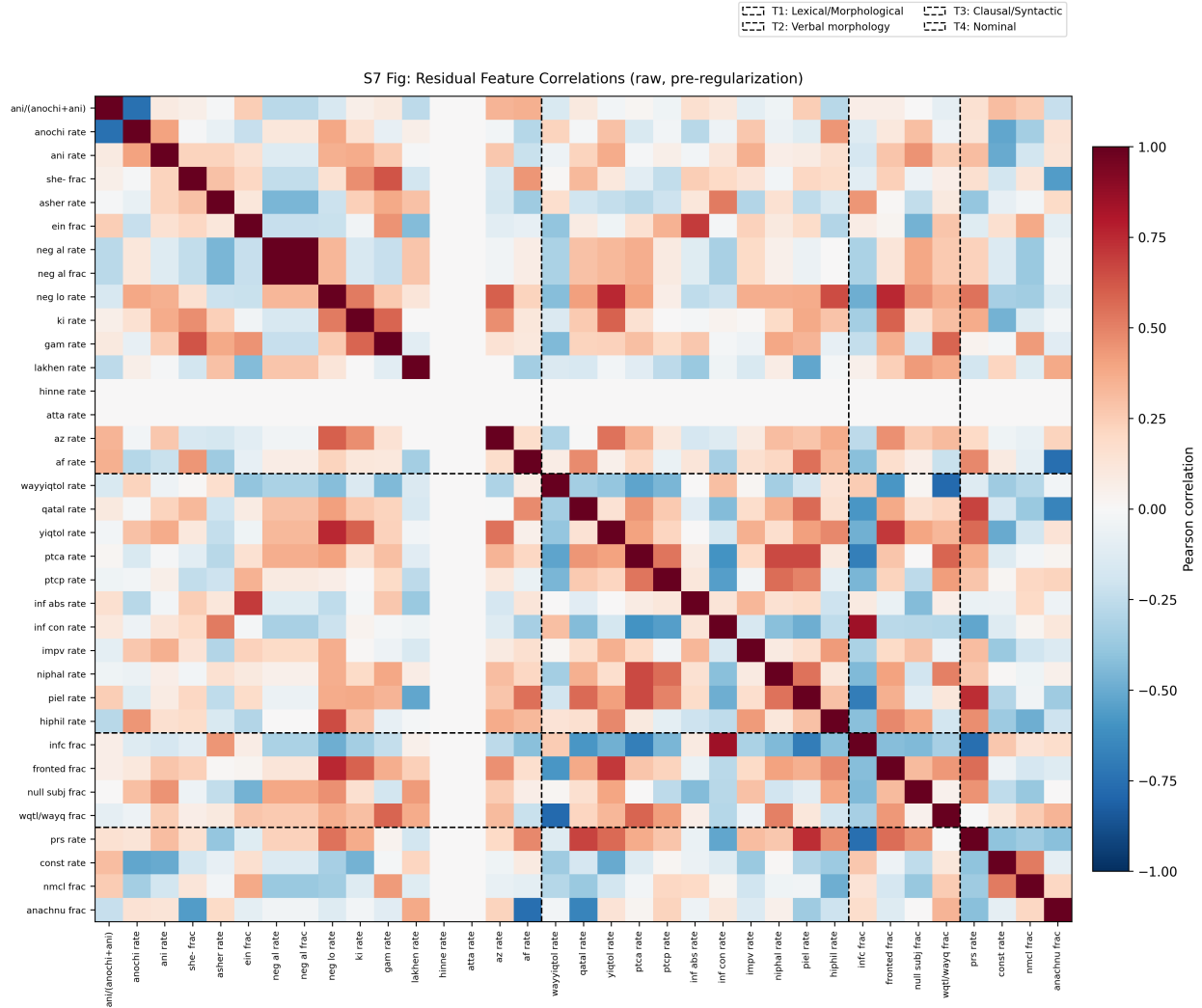


Fig 6. S7 Fig. Raw residual feature correlations (pre-regularisation). Pearson correlations among OLS residuals across the 22 training texts, computed *before* Tikhonov regularisation ($\lambda = 0.10$). Regularisation shrinks these off-diagonal entries toward zero to ensure the 35×35 covariance matrix Σ is invertible (the raw matrix is rank-deficient because $N = 22 < J = 35$). Tier boundaries are marked by dashed lines. Within-tier correlations—particularly the negative coupling between complementary forms (e.g., *wayyiqtol/qatal*, *asher/she-*)—confirm that features co-vary beyond the linear time trend, motivating the full MVN likelihood over an independent-feature approximation.

Table 7. S5 Table. Token counts for all units. Counts are from the BHSA extraction used for every feature rate in this paper. Source composites overlap the books that contain them and are not independent; D source and Deuteronomy are the same text.

Unit	Words	Role
Chronicles	35,330	training
Ezekiel	26,182	training
Jeremiah oracle	14,539	holdout
Isaiah 1–39	13,504	training
Nehemiah	7,842	training
Isaiah 40–55	5,738	training
Ezra	5,268	training
Esther	4,621	training
Ecclesiastes	4,233	training
Isaiah 56–66	3,689	training
Daniel (Heb.)	3,437	training
Hosea	3,146	training
Amos	2,780	training
Zechariah 1–8	2,484	training
Zechariah 9–14	1,987	training
Lamentations	1,945	training
Micah	1,895	training
Joel	1,318	training
Malachi	1,187	training
Zephaniah	1,037	training
Jonah	985	training
Habakkuk	897	training
Haggai	877	holdout
Nahum	746	training
Obadiah	392	training
P source	54,435	target
JE source	37,752	target
Genesis	28,764	target
Exodus	23,748	target
Numbers	23,188	target
Genesis JE	22,409	target
Deuteronomy	20,128	target
D source	20,128	target
Leviticus	17,099	target
Jeremiah Dtr	15,197	target
D Frame	11,993	target
Exodus JE	11,780	target
Leviticus P	10,496	target
D Code	7,353	target
Holiness Code	5,958	target
Numbers JE	3,563	target
D Song	782	target
Song of Deborah	462	target
Song of the Sea	433	target